

Avirup Banik

Full-Stack Software Engineer

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SUMMARY

Full-stack Software Engineer at Flipkart (Application Engineer 2) with 3.5 years of experience building production infrastructure platforms and developer tooling. Builds self-serve Kubernetes platforms, configuration management systems, real-time dashboards, and profiling/debugging tools used across engineering. Experienced in TypeScript, React/Next.js, Go, Node.js, Python, REST APIs, microservices, Kubernetes, Docker, and GCP.

EXPERIENCE

Application Engineer 2

Flipkart

Mar 2025 – Present
Bengaluru, India *Promoted to full-time after 2 years as a contractor on the same team.*

- Built a self-service Kubernetes platform replacing ticket-driven operations for namespace provisioning, CPU/memory quotas, cross-team resource requests, and quota reclamation across Flipkart datacenter clusters, reducing on-call workload by 50%.
- Owned the React/TypeScript frontend and Kubernetes-backed APIs for a configuration management platform used by 50+ microservice teams, enabling production configuration changes without application redeployments.
- Built Flipkart's internal profiling and debugging platform with a 3-person team using React and Go, providing flame graphs, profile comparison, and GC analysis for high-traffic production services.
- Extended the platform with host-level diagnostics, including TCP packet capture, expanding support beyond Java services and reducing reliance on external debugging tooling.
- Owned 4 backend microservices across Go and Node.js; optimized MySQL queries and introduced automated database migrations, eliminating manual schema changes.
- Built a real-time cluster workload dashboard using Server-Sent Events, REST, Prometheus, and Grafana, reducing production investigation time by 30%; drove RCA-based fixes that reduced repeat incidents by 25%.
- Mentored new joiners and conducted technical interviews for intern hiring.

SDE-UI

Flipkart (via Thinkify Labs)

Apr 2023 – Mar 2025
Bengaluru, India

- Migrated a production load-balancer operations console from Angular to React and TypeScript in 8 weeks, introducing reusable components and reducing follow-on UI defects by 30%.
- Migrated a second infrastructure UI from vanilla JavaScript to Next.js and TypeScript, then expanded the platform to support capacity planning, tenant onboarding, resource hierarchy, and operations workflows used across Compute, Network Services, ALM, and Data Center teams.
- Built a Docker, Kubernetes, and Helm-based CI/CD pipeline for staging and production, reducing release time from hours to under 15 minutes.
- Increased unit and integration test coverage to 95%+ across legacy Go and JavaScript projects, improving regression coverage for critical production workflows.

PROJECTS & OPEN SOURCE

SnapSocial

Live Demo | GitHub

React, TypeScript, Cloudflare Workers, D1, R2

- Built and deployed a full-stack photo-sharing platform on Cloudflare Workers with React, D1, and R2, including client-side image processing, optimistic UI, and REST APIs.
- Implemented PBKDF2 password hashing, CSRF protection, per-route rate limiting, and an admin moderation panel.

Open Source: Contributed 4 PRs each to **Kestra** (Vue.js) and **LitmusChaos** (TypeScript, React), improving responsive layouts and fixing UI issues. Kestra | LitmusChaos

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL

Frontend: React, Next.js, Angular, Tailwind CSS, Material UI

Backend & APIs: Node.js, Express.js, REST APIs, microservices

Databases: MySQL, SQL query optimization, database migrations

Cloud & Infrastructure: Kubernetes, Docker, Helm, GCP, Linux, Nginx

Observability & Testing: Prometheus, Grafana, performance profiling, flame graphs, unit & integration testing, root-cause analysis

Engineering: Platform engineering, distributed systems, full-stack development, developer tooling, real-time systems